COEP Technological University

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A Unitary Public University of Government of Maharashtra

End Semester Examination

Program : S.Y. B.Tech. | Semester: IV
Course Code: OEC-24016 | Course Name: Complex Analysis
Branch: ALL | Academic Year: 2025-26
Duration: 120 minutes | Maximum Marks: 50
Date : 21/4/2026

Student MIS Number : | | | | | | | | | |

Instructions :

1. Write your MIS Number and Branch on the Question Paper.
2. Writing anything else on the question paper is not allowed.
3. Read a given question very carefully and then answer.
4. Mobile phones and calculators are strictly prohibited.
5. Exchange/Sharing of stationery, etc. is not allowed.
6. Figures to the right indicate the course outcomes, full marks and Learning level.
7. Supposing the paper is lengthy, it is advised to manage your time in the exam accordingly.
8. Let $\mathbb{C}$ denotes the set of complex numbers.
9. For $z \in \mathbb{C} \setminus \{0\}$, let $\arg(z)$ be chosen such that $\arg(z) \in (-\pi, \pi]$.

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1. State True or False **without justification.**

   (a) The value of
   $$\int_{|z|=1} \sin\left(\frac{1}{z^{2024}}\right) = 0.$$
   [CO3, 1 Mark, L3]

   (b) Let $w : \left\{z : 0 \leq \arg(z) \leq \frac{2\pi}{n}\right\} \to \mathbb{C}$ be given by $w(z) = z^n$ for a fixed $n > 3$. Then w can take any value from $\mathbb{C}$. [CO3, 1 Mark, L3]

   (c) The power series
   $$\sum_{n=1}^{\infty} n^5 z^n$$
   has radius of convergence equal to 1. [CO3, 1 Mark, L3]

   (d) The function
   $$\frac{(\sin z^2)(\cos z^3)\left(e^{z^8}\right)}{z^{12}}$$
   has a pole of order 10 at $z_0 = 0$. [CO3, 1 Mark, L3]

(e) Let A be a compact set in $\mathbb{C}$. Let $B \subset A$, then B is a compact set in $\mathbb{C}$. [CO2, 1 Mark, L2]

(f) Let $v = \frac{az+b}{cz+d}$, where $a, b, c, d \in \mathbb{R}$ and $z \in \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$. Then we have

$$\mathrm{Im}(v) = \frac{(ad - bc)\,\mathrm{Im}(z)}{|cz + d|^2}.$$

[CO3, 1 Mark, L3]

2. (a) State residue theorem. [CO1, 2 Marks, L2]

(b) With justification, evaluate the integral

$$\oint_C \frac{16z - z^2 + z^4}{(z - i)^3}\, dz,$$

where C is the circle,

(i) $|z| = 2$ [CO3, 2 Marks, L3]

(ii) $|z + i| = a$, where $a \in \mathbb{R}$ and $0 < a < 2$. [CO2, 2 Marks, L2]

3. Let $A \subset \mathbb{C}$ and A be compact. Let $f : A \to \mathbb{C}$ be complex differentiable/analytic on A. Recall that $f(A) = \{f(z) : z \in A\}$ i.e. $f(A)$ is the image of f over the domain A. Does there exists A and f such that $f(A)$ is open in $\mathbb{C}$? Jusify. [CO4, 3 Marks, L4]

4. Let the function g be the pointwise limit of a power series in some neighborhood of the origin.

(a) Write down the Maclaurin series expansion of g' and g''. [CO3, 2 Marks, L3]

(b) Let z be such that $|z| < 1$. Write the first four terms for the Maculaurin series expansion of $\frac{1}{(1-z)^3}$. [CO3, 2 Marks, L3]

(c) Evaluate the sum

$$\frac{12}{100} + \frac{20}{1000} + \frac{30}{10000} + \frac{42}{100000} + \dots \quad .$$

[CO5, 2 Marks, L3]

5. Let $D \subset \mathbb{C}$. Let f_n be a sequence of functions on D such that f_n converges pointwise to f.

(a) Find f, where

$$f_n(z) = \frac{1}{1 + nz^2}$$

for $|z| \leq 1$. [CO3, 1 Marks, L3]

(b) Find $\lim_{z\to 0} (\lim_{n\to\infty} f_n(z))$ and $\lim_{n\to\infty} (\lim_{z\to 0} f_n(z))$. Justify if f_n converges uniformly to f or not. [CO3, 3 Marks, L3]

(c) Let g_n be a sequence of analytic functions on a domain D converging to g. Is g analytic on D? Explain. [CO4, 2 Marks, L4]

[P.T.O]

6. Similar to the definition of $\sin z, \cos z$, we can define

$$\cosh z = \frac{e^z + e^{-z}}{2} \quad \text{and} \quad \sinh z = \frac{e^z - e^{-z}}{2}.$$

(a) Find complex numbers z such that $\sinh z = 4$. [CO3, 2 Marks, L3]

OR

Find complex numbers z such that $\cosh z = 4$.

(b) Prove that $|\sin z|^2 = \sin^2 x + \sinh^2 y$ where $z = x + iy$ with $x, y \in \mathbb{R}$. [CO3, 2 Marks, L3]

(c) Write first two terms Laurent expansion of

$$\frac{(\sinh z^3)^2}{z^7}$$

at 0. [CO3, 2 Marks, L3]

(d) Find

$$\int_{|z|=2} \frac{(\sinh z^3)^2}{z^7}\, dz.$$

[CO3, 1 Marks, L3]

7. (a) Let f be analytic on a domain D. Write down Cauchy Riemann equations for cartesian coordinates and polar coordinates. [CO3, 2 Marks, L3]

(b) Let $h(z) = |z|^2 + z^3$, where $z \in \mathbb{C}$. Find all $z \in \mathbb{C}$ where f does not satisfy the Cauchy Riemann equation. Find all points where f is complex differentiable. [CO3, 3 Marks, L3]

OR

Let $h(z) = |z|^3 + z^2$, where $z \in \mathbb{C}$. Find all $z \in \mathbb{C}$ where f does not satisfy the Cauchy Riemann equation. Find all points where f is complex differentiable.

8. Let $\alpha > 1$ and $n \in \mathbb{N}$.

(a) Find the value of

$$\int_{|z|=1} \frac{z^n}{z^2 + 2\alpha z + 1}.$$

[CO3, 2 Marks, L3]

(b) Find the value of

$$\int_0^{2\pi} \frac{\cos(15\theta)}{\cos\theta + \sqrt{5}}\, d\theta.$$

[CO4, 3 Marks, L4]

(c) If f has a pole of order k at $z = z_0$, then write down the formula to find residue at z_0 which we denote by $\text{Res}[f(z); z_0]$. Find

$$\text{Res}\left[\frac{e^z \sin z}{(z-1)^3}; 1\right].$$

[CO2, 2 Marks, L2]

[P.T.O]

9. Prove Cauchy's inequality given as follows. Suppose f is analytic inside and on the circle C having centre at z_0 and radius r. If $|f(z)| \leq M$ on C, then

$$|f^{(n)}(z_0)| \leq \frac{n!M}{r^n}, \quad n = 1, 2, \ldots \quad .$$

Give an example of a real-differentiable function $g : (0, \infty) \to \mathbb{R}$ which does not satisfying Cauchy's inequality. [CO5, 4 Marks, L3]

OR

(a) Prove Liouville's theorem given as follows. An entire bounded complex valued function is always a constant function. [CO5, 2 Marks, L2]

(b) Let g be an entire function such that imaginary part of $g(z)$ is always less than 1 for all $z \in \mathbb{C}$. Find all possibilities of g. [CO5, 2 Marks, L2]

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1(a) **True.** The function $f(z) = \sin(z^{-2024})$ has an isolated singularity at $z = 0$. Its Laurent series expansion is $\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!z^{2024(2k+1)}}$. The coefficient of z^{-1} is zero, so by the Residue Theorem,

$$\int_{|z|=1} \sin(z^{-2024})dz = 2\pi i \cdot \mathrm{Res}(f, 0) = 0.$$

(b) **True.** The mapping $w(z) = z^n$ for z in the sector $S = z : 0 \le \arg(z) \le \frac{2\pi}{n}$ maps the boundary rays $[0, \infty)$ and $re^{i2\pi/n} : r \ge 0$ to the same ray $[0, \infty)$ in the w-plane, effectively covering the entire complex plane.

(c) **True.** Let $a_n = n^5$. By the Cauchy-Hadamard formula, the radius of convergence R is given by $1/R = \limsup_{n\to\infty} |a_n|^{1/n} = \lim_{n\to\infty}(n^5)^{1/n} = 1^5 = 1$. Thus, $R = 1$.

(d) **True.** Expanding the numerator: $\sin(z^2) = z^2 - \frac{z^6}{6} + \frac{z^{10}}{120} - \ldots$, $\cos(z^3) = 1 - \frac{z^6}{2} + \ldots$, and $e^{z^8} = 1 + z^8 + \ldots$. The product is $(z^2 - \frac{z^6}{6} + \ldots)(1 - \frac{z^6}{2} + \ldots)(1 + z^8 + \ldots) = z^2 - \frac{z^6}{6} - \frac{z^8}{2} + \ldots$. Dividing by z^{12} yields $z^{-10} - \frac{1}{6}z^{-6} - \frac{1}{2}z^{-4} + \ldots$, confirming a pole of order 10 at $z = 0$.

(e) **False.** The statement: Let A be a compact set, then $B \subset A$ is a compact set is false. Take $A = \{z : |z| \le 1\}, B = \{z : |z| < 1\}$.

(f) **True.** For $v = \frac{az+b}{cz+d}$, we have $v = \frac{(az+b)(c\bar{z}+d)}{|cz+d|^2}$. The imaginary part of the numerator $(az+b)(c\bar{z}+d)$ is $\mathrm{Im}(ac|z|^2 + adz + bc\bar{z} + bd) = ad\mathrm{Im}(z) - bc\mathrm{Im}(z) = (ad - bc)\mathrm{Im}(z)$. Thus, $\mathrm{Im}(v) = \frac{(ad-bc)\mathrm{Im}(z)}{|cz+d|^2}$.

2 (a) Statement of the residue theorem: Let f be a function that is analytic on a simply connected domain U except for a finite number of isolated singularities $z_1, z_2, \ldots, z_n$ inside U. Let C be a positively oriented, simple closed contour in U such that C encloses the points $z_1, \ldots, z_n$ and no other singularities. Then

$$\oint_C f(z)\,dz = 2\pi i \sum_{k=1}^{n} \mathrm{Res}(f, z_k),$$

where $\mathrm{Res}(f, z_k)$ denotes the residue of f at z_k.

(b) Evaluation of the integral

$$\oint_C \frac{16z - z^2 + z^4}{(z-i)^3}\,dz$$

on the given contours.

Define

$$f(z) = \frac{16z - z^2 + z^4}{(z-i)^3}.$$

The only singularity of f is at $z = i$ (a pole of order 3). The residue theorem therefore tells us that

$$\oint_C f(z)\,dz = \begin{cases} 2\pi i \cdot \mathrm{Res}(f, i) & \text{if } C \text{ encloses } z = i, \\ 0 & \text{otherwise.} \end{cases}$$

(i) Contour $C : |z| = 2$.

The circle $|z| = 2$ is centered at $z = 0$ with radius 2. The point $z = i$ has $|i| = 1 < 2$, so $z = i$ lies *inside* the contour. We must compute the residue at $z = i$.

Since $f(z)$ has a pole of order 3 at $z = i$, the residue is given by

$$\operatorname{Res}(f, i) = \frac{1}{2!} \lim_{z \to i} \frac{d^2}{dz^2} \left[(z-i)^3 f(z) \right] = \frac{1}{2} \lim_{z \to i} \frac{d^2}{dz^2} \left(16z - z^2 + z^4 \right).$$

Compute the derivatives:

$$g(z) = 16z - z^2 + z^4, \quad g'(z) = 16 - 2z + 4z^3, \quad g''(z) = -2 + 12z^2.$$

So

$$\operatorname{Res}(f, i) = \frac{1}{2} g''(i) = \frac{1}{2} \left(-2 + 12(i)^2 \right) = \frac{1}{2} \left(-2 + 12(-1) \right) = \frac{1}{2}(-14) = -7.$$

Therefore,

$$\oint_{|z|=2} \frac{16z - z^2 + z^4}{(z-i)^3} \, dz = 2\pi i \cdot (-7) = -14\pi i.$$

(ii) Contour $C : |z + i| = a$, with $a \in \mathbb{R}$, $0 < a < 2$.

The circle $|z+i| = a$ is centered at $z = -i$ with radius a. The singularity is at $z = i$. The distance between the center $-i$ and the point i is

$$|i - (-i)| = |2i| = 2.$$

Since $0 < a < 2$, the radius a is strictly less than 2, so the point $z = i$ lies *outside* the disk $|z + i| < a$, and hence outside the contour C.

Inside C, the function

$$f(z) = \frac{16z - z^2 + z^4}{(z-i)^3}$$

is holomorphic (no singularities inside or on C), so by Cauchy's integral theorem,

$$\oint_{|z+i|=a} f(z) \, dz = 0.$$

3. Since $f : A \to \mathbb{C}$ is analytic, it is continuous. The continuous image of a compact set is compact, so $f(A)$ is compact. In $\mathbb{C}$, compact sets are closed and bounded. Suppose $f(A)$ is open. Then $f(A)$ is both open and closed in $\mathbb{C}$. Since $f(A)$ can not be empty, $f(A)$ must be $\mathbb{C}$. However $f(A)$ being bounded would imply $\mathbb{C}$ bounded, an absurdity. Thus, no such non-trivial A and f exist.

4(a) The Maclaurin series expansion of $g(x)$ is

$$g(x) = g(0) + g'(0)x + \frac{g''(0)}{2!}x^2 + \frac{g'''(0)}{3!}x^3 + \cdots.$$

Differentiating term-by-term gives the Maclaurin series for $g'(x)$:

$$g'(x) = g'(0) + g''(0)x + \frac{g'''(0)}{2!}x^2 + \frac{g^{(4)}(0)}{3!}x^3 + \cdots.$$

Differentiating again yields the series for $g''(x)$:

$$g''(x) = g''(0) + g'''(0)x + \frac{g^{(4)}(0)}{2!}x^2 + \frac{g^{(5)}(0)}{3!}x^3 + \cdots.$$

4(b) Differentiating the geometric series $\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$ $(|z| < 1)$ twice gives

$$\frac{1}{(1-z)^3} = \sum_{n=0}^{\infty} \frac{(n+1)(n+2)}{2} z^n.$$

The first four terms ($n = 0$ to $n = 3$) are

$$1 + 3z + 6z^2 + 10z^3.$$

4(c) The general term is $\frac{(n+2)(n+1)}{10^{n+2}}$ for $n \geq 0$. Thus,

$$S = \sum_{n=0}^{\infty} \frac{(n+2)(n+1)}{10^{n+2}} = \frac{1}{100} \sum_{n=0}^{\infty} (n+2)(n+1) \left(\frac{1}{10}\right)^n.$$

From part (ii), $\sum_{n=0}^{\infty} (n+2)(n+1)w^n/2 = \frac{1}{(1-w)^3}$ $(|w| < 1)$, so with $w = 1/10$,

$$\sum_{n=0}^{\infty} (n+2)(n+1) \left(\frac{1}{10}\right)^n = 2 \cdot \frac{1}{(9/10)^3} = \frac{2 \cdot 1000}{729} = \frac{2000}{729}.$$

Hence,

$$S = \frac{1}{100} \cdot \frac{2000}{729} = \frac{20}{729}.$$

5 (a) Pointwise limit of $f_n(z)$:
For $z = 0$, $f_n(0) = \frac{1}{1+0} = 1$ for all n, so $\lim_{n\to\infty} f_n(0) = 1$. For $z \neq 0$ such that $|z| \leq 1$, as $n \to \infty$, $nz^2 \to \infty$, so $f_n(z) = \frac{1}{1+nz^2} \to 0$. Thus, the pointwise limit $f(z)$ is:

$$f(z) = \begin{cases} 1 & \text{if } z = 0 \\ 0 & \text{if } 0 < |z| \leq 1 \end{cases}$$

(b) Limit interchange and uniform convergence: Let $\lim_{n\to\infty} f_n(z) = f(z)$. Therefore,

$$\lim_{z\to 0}(\lim_{n\to\infty} f_n(z)) = \lim_{z\to 0} f(z).$$

Since $f(z) = 0$ for $z \neq 0$, the limit as $z \to 0$ is 0. Second, $\lim_{z\to 0} f_n(z) = \frac{1}{1+n(0)^2} = 1$. Consequently, $\lim_{n\to\infty}(\lim_{z\to 0} f_n(z)) = \lim_{n\to\infty}(1) = 1$. The convergence is **not uniform**. If a sequence of continuous functions f_n converges uniformly to a limit f, then f must be continuous. Since each f_n is continuous but the limit function f is discontinuous at $z = 0$, the convergence cannot be uniform.
(c) Analyticity of the limit: The answer is **NO**. If a sequence of analytic functions $\{g_n\}$ on a domain D converges **uniformly on compact subsets** of D to a function g, then g is indeed analytic on D (This was done in class). However, if the sequence only converges pointwise, the limit function g is not necessarily analytic or even continuous.

- **6 (a) Solutions for $\sinh z = 4$ or $\cosh z = 4$:**

  For $\sinh z = 4$: $\frac{e^z - e^{-z}}{2} = 4 \implies e^z - e^{-z} = 8$. Let $u = e^z$, then $u - \frac{1}{u} = 8 \implies u^2 - 8u - 1 = 0$. Using the quadratic formula, $u = \frac{8 \pm \sqrt{64+4}}{2} = 4 \pm \sqrt{17}$. $z = \ln(4 \pm \sqrt{17}) + 2k\pi i$, for $k \in \mathbb{Z}$.

  For $\cosh z = 4$: $\frac{e^z + e^{-z}}{2} = 4 \implies e^z + e^{-z} = 8$. Let $u = e^z$, then $u^2 - 8u + 1 = 0$. $u = \frac{8 \pm \sqrt{64-4}}{2} = 4 \pm \sqrt{15}$. $z = \ln(4 \pm \sqrt{15}) + 2k\pi i$, for $k \in \mathbb{Z}$.

- **(b) Proof that $|\sin z|^2 = \sin^2 x + \sinh^2 y$:**
  Note that $\cosh^2 y = 1 + \sinh^2 y$ for $y \in \mathbb{R}$. Using the identity

$$\sin(x+iy) = \sin x \cos iy + \cos x \sin iy = \sin x \cosh y + i \cos x \sinh y,$$

  we have,

$$|\sin z|^2 = (\sin x \cosh y)^2 + (\cos x \sinh y)^2$$
$$= \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y = \sin^2 x(1+\sinh^2 y) + (1-\sin^2 x)\sinh^2 y$$
$$= \sin^2 x + \sin^2 x \sinh^2 y + \sinh^2 y - \sin^2 x \sinh^2 y = \sin^2 x + \sinh^2 y.$$

- **(c) First two terms of the Laurent expansion of $\frac{(\sinh z^3)^2}{z^7}$ at 0:**
  We have $\sinh u = \frac{e^u - e^{-u}}{2} = u + \frac{u^3}{6} + \dots$. Then

$$\sinh z^3 = z^3 + \frac{(z^3)^3}{6} + \dots = z^3 + \frac{z^9}{6} + \dots.$$

  Squaring this:

$$(\sinh z^3)^2 = (z^3 + \frac{z^9}{6} + \dots)^2 = z^6 + 2 \cdot z^3 \cdot \frac{z^9}{6} + \dots = z^6 + \frac{z^{12}}{3} + \dots.$$

  Dividing by z^7: $f(z) = \frac{z^6 + \frac{z^{12}}{3} + \dots}{z^7} = \frac{1}{z} + \frac{z^5}{3} + \dots$. The first two terms are $\frac{1}{z} + \frac{z^5}{3}$.

- **(d) Value of**

$$\int_{|z|=2} \frac{(\sinh z^3)^2}{z^7}\,dz:$$

  Using the Cauchy Residue Theorem, the integral is $2\pi i \cdot \mathrm{Res}(f,0)$. From the Laurent expansion in the previous step, $f(z) = \frac{1}{z} + \dots$, so the residue at $z=0$ is 1. Therefore,

$$\int_{|z|=2} f(z)dz = 2\pi i(1) = 2\pi i.$$

7 (a) **Cauchy–Riemann equations**

Let

$$f(z) = u(x,y) + iv(x,y), \qquad z = x + iy.$$

Then the Cauchy–Riemann equations in **Cartesian coordinates** are

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

In **polar coordinates**, where

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

and $f(z) = u(r,\theta) + iv(r,\theta)$, the Cauchy–Riemann equations are

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \qquad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}, \qquad (r \neq 0).$$

(b) **For** $h(z) = |z|^2 + z^3$

Write $z = x + iy$. Then

$$|z|^2 = x^2 + y^2, \qquad z^3 = (x+iy)^3.$$

Hence

$$h(z) = \big(x^2 + y^2 + x^3 - 3xy^2\big) + i\big(3x^2y - y^3\big).$$

So,

$$u(x,y) = x^2 + y^2 + x^3 - 3xy^2, \qquad v(x,y) = 3x^2y - y^3.$$

Compute the partial derivatives:

$$u_x = 2x + 3x^2 - 3y^2, \qquad u_y = 2y - 6xy,$$

$$v_x = 6xy, \qquad v_y = 3x^2 - 3y^2.$$

The Cauchy–Riemann equations require

$$u_x = v_y \quad \text{and} \quad u_y = -v_x.$$

Therefore,

$$2x + 3x^2 - 3y^2 = 3x^2 - 3y^2 \implies x = 0,$$

and

$$2y - 6xy = -6xy \implies y = 0.$$

Thus the Cauchy–Riemann equations are satisfied only at

$$z = 0.$$

Since $h(z)$ is a polynomial in z and $\bar{z}$, it is complex differentiable only where the Cauchy–Riemann equations hold. Hence h is complex differentiable only at

$$z = 0.$$

Answer: $h(z)$ does not satisfy the Cauchy–Riemann equations at every point except $z = 0$, and it is complex differentiable only at $z = 0$.

OR

$$\text{Let } h(z) = |z|^3 + z^2, \qquad z \in \mathbb{C}.$$

Write $z = x + iy$. Then

$$|z|^3 = (x^2 + y^2)^{3/2}, \qquad z^2 = (x^2 - y^2) + i(2xy).$$

So

$$h(z) = u(x,y) + iv(x,y),$$

where

$$u(x,y) = (x^2 + y^2)^{3/2} + x^2 - y^2, \qquad v(x,y) = 2xy.$$

Now compute the partial derivatives:

$$u_x = 3x\sqrt{x^2 + y^2} + 2x, \qquad u_y = 3y\sqrt{x^2 + y^2} - 2y,$$

$$v_x = 2y, \qquad v_y = 2x.$$

The Cauchy–Riemann equations are

$$u_x = v_y, \qquad u_y = -v_x.$$

Hence

$$3x\sqrt{x^2+y^2} + 2x = 2x,$$

so

$$3x\sqrt{x^2+y^2} = 0,$$

and

$$3y\sqrt{x^2+y^2} - 2y = -2y,$$

so

$$3y\sqrt{x^2+y^2} = 0.$$

Therefore,

$$x = 0, \qquad y = 0.$$

Thus the Cauchy–Riemann equations are satisfied only at

$$z = 0.$$

So h does not satisfy the Cauchy–Riemann equations at every point $z \neq 0$.

Since the partial derivatives exist, h is complex differentiable only at the point where the Cauchy–Riemann equations hold. Hence h is complex differentiable only at

$$\boxed{z = 0.}$$

$$\boxed{h \text{ fails the Cauchy–Riemann equations for all } z \neq 0, \text{ and is complex differentiable only at } z = 0.}$$

8(a) Consider the following.

$$f(z) = \frac{z^n}{z^2 + 2\alpha z + 1} = \frac{z^n}{(z - z_1)(z - z_2)}$$

with

$$z_1 = -\alpha + \sqrt{\alpha^2 - 1} \quad \text{and} \quad z_2 = -\alpha - \sqrt{\alpha^2 - 1}.$$

Since $z_1 z_2 = 1$ and $\alpha > 1$, we have $|z_1| < 1$ and $|z_2| > 1$. It follows that the only singularity of $f(z)$ inside the unit circle $|z| = 1$ is $z = z_1$, which is a simple pole with

$$\operatorname{Res}[f(z); z_1] = \lim_{z \to z_1} (z - z_1) f(z) = \lim_{z \to z_1} \frac{z^n}{z - z_2} = \frac{z_1^n}{z_1 - z_2} = \frac{z_1^n}{2\sqrt{\alpha^2 - 1}}.$$

Therefore,

$$J = 2\pi i \operatorname{Res}[f(z); z_1] = 2\pi i \left(\frac{(\sqrt{\alpha^2 - 1} - \alpha)^n}{2\sqrt{\alpha^2 - 1}} \right) \quad \text{for } \alpha > 1.$$

(b)

$$I = \int_0^{2\pi} \frac{\cos n\theta}{\cos\theta + \alpha}\, d\theta, \quad \text{for } \alpha > 1 \text{ and } n \in \mathbb{N}_0.$$

We may rewrite the integral as

$$I = \text{Re}\left[\int_0^{2\pi} \frac{e^{in\theta}}{\cos\theta + \alpha}\, d\theta\right] = \text{Re}[J].$$

The substitution $z = e^{i\theta}$ gives

$$J = \int_{|z|=1} \frac{z^n}{\left(\frac{z^2+1}{2z}\right) + \alpha} \frac{dz}{iz} = \frac{2}{i}\int_{|z|=1} f(z)\, dz,$$

where

$$f(z) = \frac{z^n}{z^2 + 2\alpha z + 1} = \frac{z^n}{(z - z_1)(z - z_2)}$$

with

$$z_1 = -\alpha + \sqrt{\alpha^2 - 1} \quad \text{and} \quad z_2 = -\alpha - \sqrt{\alpha^2 - 1}.$$

Since $z_1 z_2 = 1$ and $\alpha > 1$, we have $|z_1| < 1$ and $|z_2| > 1$. It follows that the only singularity of $f(z)$ inside the unit circle $|z| = 1$ is $z = z_1$, which is a simple pole with

$$\text{Res}[f(z); z_1] = \lim_{z\to z_1} (z - z_1) f(z) = \lim_{z\to z_1} \frac{z^n}{z - z_2} = \frac{z_1^n}{z_1 - z_2} = \frac{z_1^n}{2\sqrt{\alpha^2 - 1}}.$$

Therefore,

$$J = \frac{2}{i}(2\pi i \text{Res}[f(z); z_1]) = 2\pi\left(\frac{(\sqrt{\alpha^2 - 1} - \alpha)^n}{\sqrt{\alpha^2 - 1}}\right) \quad \text{for } \alpha > 1.$$

Consequently (note that $\text{Im}\, J = 0$), $I = J$. For instance, for $n = 1$ and $\alpha = 2$, one has

$$\int_0^{2\pi} \frac{\cos(15\theta)}{\cos\theta + \sqrt{5}}\, d\theta = 2\pi\left(\frac{(2 - \sqrt{5})^{15}}{2}\right).$$

(c) If $f(z)$ has a pole of order k at $z = z_0$, the residue is given by the formula:

$$\text{Res}[f(z); z_0] = \frac{1}{(k-1)!} \lim_{z\to z_0} \frac{d^{k-1}}{dz^{k-1}} \left[(z - z_0)^k f(z)\right].$$

For $f(z) = \frac{e^z \sin z}{(z-1)^3}$, the pole is of order $k = 3$ at $z_0 = 1$. The residue is:

$$\text{Res}\left[\frac{e^z \sin z}{(z-1)^3}; 1\right] = \frac{1}{2!} \lim_{z\to 1} \frac{d^2}{dz^2}(e^z \sin z).$$

To evaluate this, we compute the derivatives of $g(z) = e^z \sin z$:

$$g'(z) = e^z \sin z + e^z \cos z = e^z(\sin z + \cos z),$$

$$g''(z) = e^z(\sin z + \cos z) + e^z(\cos z - \sin z) = 2e^z \cos z.$$

Substituting $z = 1$ into the residue formula:

$$\text{Res}\left[\frac{e^z \sin z}{(z-1)^3}; 1\right] = \frac{1}{2}(2e^1 \cos(1)) = e\cos(1).$$

9. By the generalised Cauchy integral formula ,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{n+1}}\,dz.$$

Hence, by the hypothesis and ML inequality

$$|f^{(n)}(z_0)| \leq \frac{n!}{2\pi}\int_C \frac{|f(z)|}{|z-z_0|^{n+1}}|dz| \leq \frac{n!M}{2\pi}\int_C \frac{|dz|}{|z-z_0|^{n+1}}.$$

Since $|z-z_0| = r$,

$$\frac{n!M}{2\pi}\int_C \frac{|dz|}{|z-z_0|^{n+1}} = \frac{n!M}{2\pi}\int_C \frac{|dz|}{r^{n+1}} = \frac{n!M}{r^n}.$$

Consider $f(x) = \cos(1/x)$ for $x > 0$. Then f is differentiable for $x > 0$ and $|f(x)| \leq 1$ for all $x > 0$. However,

$$f'(x) = \frac{1}{x^2}\sin(1/x)$$

which is clearly not bounded. To see this for each $n \in \mathbb{N}$, consider

$$\left|f'\left(\frac{1}{n\pi+\frac{\pi}{2}}\right)\right| = \left(n\pi + \frac{\pi}{2}\right)^2.$$

As $n \to \infty$, $\left|f'\left(\frac{1}{n\pi+\frac{\pi}{2}}\right)\right| \to \infty$.